

Centre for infrastructure, Sustainable Transportation and Urban Planning

Indian Institute of Science (IISc), Bengaluru

Follow the instructions below precisely.

- Plagiarism will result in instant disqualification. You must submit your own submission.
- You can make your submission here: <https://forms.gle/uL9PisZtUxsU4Zjw8>
- Please take the test only if you can commit full-time for 3+ months.
- You are allowed to make only one submission.
- Last Date to submit your work: 22nd July 2025 (05:00 PM)
- Late submissions will not be accepted.
- If selected, it will be mandatory for you to join in an in-person capacity. Please refrain from attempting the test if you cannot attend in-person.
- As part of the internship selection process you will be required to develop a Power BI dashboard.
- The test will be used to shortlist for “PowerBI intern”, supervised by Dr. Vijay Gopal Kovvali and Dileepan K.
- For queries, contact Dileepan K (dileepank@iisc.ac.in).

All the best!

CiSTUP

Indian Institute of Science, Bengaluru

Question

Your task is to create a comprehensive Power BI dashboard analyzing road safety data. This assignment aims to evaluate your skills in data analysis, visualization, and dashboard creation using Power BI. The dataset provided contains information about road accidents, including various attributes such as accident date, severity, weather conditions, and more. You are expected to explore this dataset, identify key performance indicators (KPIs), and present insightful visualizations through a well-designed dashboard.

Dataset Link: https://drive.google.com/file/d/1yTI0506e7lZglRI7fsG2_TL0fHcd-G6D/view

1 Data Preparation:

- Import the provided dataset into Power BI.
- Clean the data by handling missing values, outliers, and any inconsistencies.
- Perform necessary data transformations to prepare it for analysis.

2 Exploratory Data Analysis (EDA):

- Explore the dataset to understand the distribution and characteristics of different variables.
- Identify trends, patterns, and correlations within the data.

- Select relevant KPIs based on your analysis.

3. Dashboard Design:

- Design an interactive dashboard with multiple visualizations to effectively communicate insights from the data.
- Include a variety of charts and graphs such as bar charts, line charts, maps, slicers etc.
- Ensure the dashboard layout is intuitive and user-friendly.

4. Key Performance Indicators (KPIs):

Choose KPIs that provide meaningful insights into road safety and accident trends. Examples of KPIs include:

- Accident Severity Distribution
- Number of Accidents Over Time
- Distribution of Accidents by Day of the Week
- Top Junction Controls Contributing to Accidents
- Weather Conditions vs. Accident Severity
- Road Surface Conditions vs. Accident Severity

Submission Guidelines:

- Submit your Power BI dashboard file (.pbix) along with a summary document (.pdf) highlighting your analysis approach, key findings, and insights derived from the dashboard.
- Ensure your dashboard is well-documented and easy to navigate.
- Upload your files (.pbix and .pdf) on google drive and under the General Access tab select 'Anyone with link'.